

Rotations in $\mathbb{R}^2$

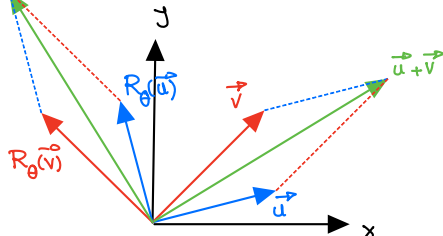
Definition

The transformation $R_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ denotes counterclockwise rotation about the origin through an angle of θ .

Question

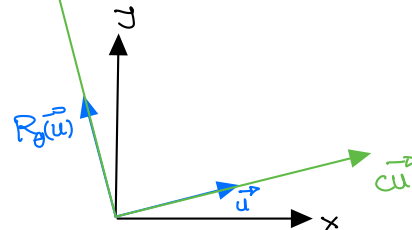
Is $R_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ a linear transformation?

$$R_\theta(\vec{u} + \vec{v}) = R_\theta(\vec{u}) + R_\theta(\vec{v})$$

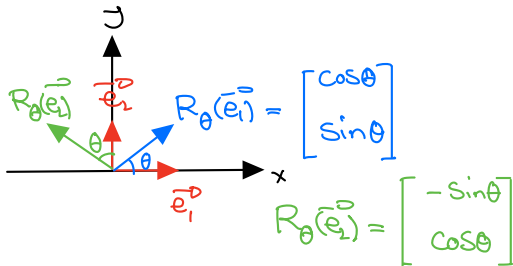


R_θ preserves vector addition.

$$R_\theta(c\vec{u}) = c R_\theta(\vec{u})$$



R_θ preserves scalar multiplication.



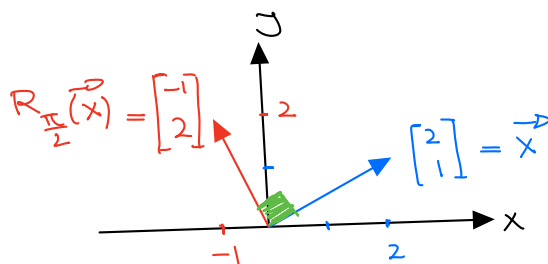
$$A_{R_\theta} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Problem

The transformation $R_{\frac{\pi}{2}} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ denotes a *counterclockwise* rotation about the origin through an angle of $\frac{\pi}{2}$ radians. Find the matrix of $R_{\frac{\pi}{2}}$.

$$A_{R_{\frac{\pi}{2}}} = \begin{bmatrix} \cos \frac{\pi}{2} & -\sin \frac{\pi}{2} \\ \sin \frac{\pi}{2} & \cos \frac{\pi}{2} \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$A_{R_{\frac{\pi}{2}}} \vec{x} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$

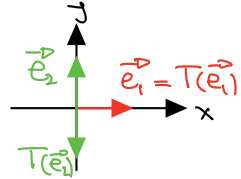


Dot product is zero: $\vec{x} \cdot R_{\frac{\pi}{2}}(\vec{x}) = 0$

Reflection in $\mathbb{R}^2$

Example

In $\mathbb{R}^2$, reflection in the x -axis, which transforms $\begin{bmatrix} a \\ b \end{bmatrix}$ to $\begin{bmatrix} a \\ -b \end{bmatrix}$, is a matrix transformation:

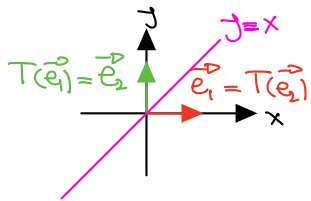


$$\begin{bmatrix} a \\ b \end{bmatrix} = a \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{\vec{e}_1} + b \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\vec{e}_2}$$

$$T \begin{bmatrix} a \\ b \end{bmatrix} = a \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{T(\vec{e}_1)} + b \underbrace{\begin{bmatrix} 0 \\ -1 \end{bmatrix}}_{T(\vec{e}_2)} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a \\ -b \end{bmatrix}$$

Example

Reflection in the line $y = x$ transforms $\begin{bmatrix} a \\ b \end{bmatrix}$ to $\begin{bmatrix} b \\ a \end{bmatrix}$. This is a matrix transformation:



$$\begin{bmatrix} a \\ b \end{bmatrix} = a \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{\vec{e}_1} + b \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\vec{e}_2}$$

$$T \begin{bmatrix} a \\ b \end{bmatrix} = a \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{T(\vec{e}_1)} + b \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{T(\vec{e}_2)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} b \\ a \end{bmatrix}$$

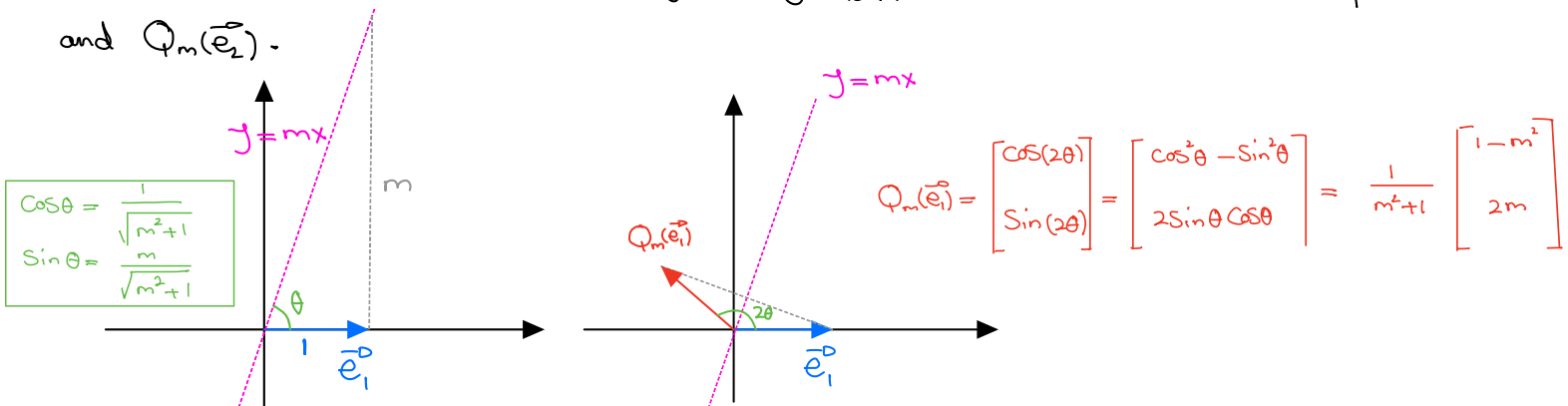
$$T_A(\vec{x}) = A \vec{x} \quad \text{where} \quad \vec{x} \in \mathbb{R}^2 \quad \text{and} \quad A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Reflection in $\mathbb{R}^2$

Problem

The reflection in the line $y = mx$ is a linear transformation. Find the matrix corresponding to this transformation.

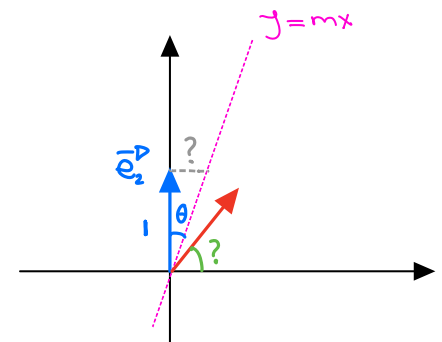
Let us denote the reflection in the line $y = mx$ by $S_m: \mathbb{R}^2 \rightarrow \mathbb{R}^2$. We need to compute $Q_m(\vec{e}_1)$ and $Q_m(\vec{e}_2)$.



Exercise: Show that $Q_m(\vec{e}_2) = \frac{1}{m^2 + 1} \begin{bmatrix} 2m \\ m^2 - 1 \end{bmatrix}$ and hence the matrix

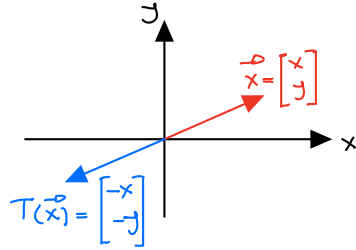
corresponding to Q_m is $\frac{1}{m^2 + 1} \begin{bmatrix} 1 - m^2 & 2m \\ 2m & m^2 - 1 \end{bmatrix}$.

Hint: First answer the two questions indicated in the picture.



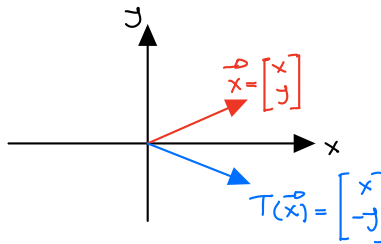
Shear, Projections, and Reflection through the origin

Reflection through the origin $\longrightarrow$ Can also be thought of as R_π .



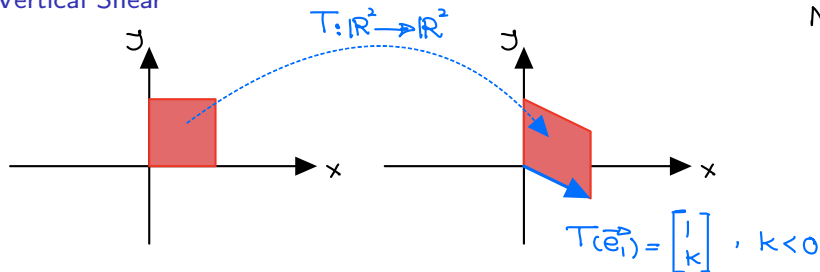
$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = R_\pi = \begin{bmatrix} \cos \pi & -\sin \pi \\ \sin \pi & \cos \pi \end{bmatrix}$$

Projection into the x -axis



$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Vertical Shear



Note: $T(\vec{e}_2) = \vec{e}_2$, $T(\vec{e}_1) = \begin{bmatrix} 1 \\ k \end{bmatrix}$ where in the figure $k < 0$.

$$A = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

Matrix Algebra

Definitions

Let m and n be positive integers.

- An $m \times n$ matrix is a rectangular array of numbers having m rows and n columns. Such a matrix is said to have size $m \times n$.
- a_{ij} is the (i, j) -entry of a matrix, which is the entry in row i and column j .

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}]$$

Other keywords: diagonal entries, the main diagonal, identity matrix, zero matrix

Example:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix} = [a_{ij}] \quad a_{23} = -1$$

the main diagonal

(Zero matrix) $0_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (Identity matrix) $I_4 = I_{4 \times 4} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Matrix addition

Definition

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two $m \times n$ matrices. Then $A + B = C$ where C is the $m \times n$ matrix $C = [c_{ij}]$ defined by

$$c_{ij} = a_{ij} + b_{ij}$$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} -1 & 2 \\ 4 & 3 \end{bmatrix}$$

$$C = A + B = \begin{bmatrix} 0 & 4 \\ 7 & 7 \end{bmatrix}$$

$$c_{12} = a_{12} + b_{12} = 2 + 2 = 4$$

Theorem (Properties of Matrix Addition)

Let A , B and C be $m \times n$ matrices. Then the following properties hold.

- ① $A + B = B + A$ (matrix addition is commutative).
- ② $(A + B) + C = A + (B + C)$ (matrix addition is associative).
- ③ There exists an $m \times n$ zero matrix, 0 , such that $A + 0 = A$. (existence of an additive identity).
- ④ There exists an $m \times n$ matrix $-A$ such that $A + (-A) = 0$. (existence of an additive inverse).

Proof of ②:

The (i, j) -entry of $(A + B) + C$ is $(a_{ij} + b_{ij}) + c_{ij}$,
which is equal to the (i, j) -entry of $A + (B + C)$, i.e., $a_{ij} + (b_{ij} + c_{ij})$.

Scalar Multiplication

Definition

Let $A = [a_{ij}]$ be an $m \times n$ matrix and let k be a scalar. Then $kA = [ka_{ij}]$.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad 2A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

Theorem (Properties of Scalar Multiplication)

Let A, B be $m \times n$ matrices and let $k, p \in \mathbb{R}$ (scalars). Then the following properties hold.

- ① $k(A + B) = kA + kB$. (scalar multiplication distributes over matrix addition).
- ② $(k + p)A = kA + pA$. (addition distributes over scalar multiplication).
- ③ $k(pA) = (kp)A$. (scalar multiplication is associative).

$$3 \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} -3 & 0 \\ 3 & -3 \end{bmatrix} = (3 - 1 + 3) \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = 5 \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ -5 & 5 \end{bmatrix}$$

Matrix Multiplication

Definition (Product of two matrices)

Let A be an $m \times n$ matrix and let B be an $n \times p$ matrix, whose columns are $B_1, B_2, \dots, B_p$. The **product of A and B** is

$$AB = A \begin{bmatrix} B_1 & B_2 & \cdots & B_p \end{bmatrix} = \begin{bmatrix} AB_1 & AB_2 & \cdots & AB_p \end{bmatrix}$$

Question: What is the size of AB ?

Problem

Find the product AB of matrices $A = \begin{bmatrix} -1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} \vec{B}_1 & \vec{B}_2 & \vec{B}_3 \\ -1 & 1 & 2 \\ 0 & -2 & 4 \\ 1 & 0 & 0 \end{bmatrix}$

$$A \vec{B}_1 = \begin{bmatrix} -1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}_{2 \times 1}$$

$$A \vec{B}_2 = \begin{bmatrix} -1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}_{2 \times 1}$$

$$A \vec{B}_3 = \begin{bmatrix} -1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}_{2 \times 1}$$

$$AB = \begin{bmatrix} 4 & -1 & -2 \\ -1 & 4 & 0 \end{bmatrix}_{2 \times 3}$$